Ahmed Fathy

Software Engineer
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Education

- **Cairo University, Faculty of Engineering** Giza, Egypt
  B.Sc. in Computer Engineering; GPA: 3.6 *2022*

Experience

- **Al Nada Scientific Office** | *Technical Support Trainee* *Cairo, Egypt | June 2022 – 2024*
  - **Diagnostics & Maintenance**: Troubleshot workstations and network equipment; performed firmware updates and preventive maintenance.
  - **Documentation**: Created systematic troubleshooting documentation to streamline hardware issue resolution.

Student Activities

- **IEEE** | *UI/UX Designer - Frontend Team Leader* *Cairo | Oct 2025 – Present*
  - **Portfolio Platform Development**: Engineered comprehensive web platform featuring workshop and event management systems with admin dashboard for content creation, role-based access control enabling instructors to upload course materials and students to register for workshops and events.
- **Technical Center for Career Development (TCCD)** | *Frontend Developer* *Cairo | June 2025 – Present*
  - **Portfolio Platform Development**: Developed comprehensive platform with event pages, admin dashboard for event creation, role-based access for instructors and students, location management, event galleries, and statistical analysis pages.
  - **HR Management & Judging System**: Built HR system managing team attendance with custom form builder and reviewer functionality similar to Google Forms, alongside event judging system for evaluating and assessing teams applying to events.
- **Physics Helping District (PhD)** | *Team Leader* *Cairo | June 2022 – 2024*
  - **Mentorship**: Led peer tutoring initiative in electronics, achieving measurable academic improvement.
- **NASA Space Apps Challenge** | *Participant – Project: "Orrery"* *Cairo | Oct 2024*
  - **Orrery App**: Developed an interactive orrery app visualizing near-Earth objects using React and Three.js.

Projects

- **CNN Convolution Accelerator** | *Verilog, OpenLane2, ASIC Design* Source Code | *2025*
  - **Architecture**: Designed a synthesizable 2D convolution accelerator using Systolic Arrays and fully pipelined dataflow to maximize MAC throughput.
  - **Memory Hierarchy**: Implemented DMA-based loading and dual-port SRAM buffering to support up to 16x16 kernel execution.
  - **Physical Design**: Executed RTL-to-GDSII synthesis using OpenLane2, validating physical feasibility and achieving timing closure.
- **AES Encryption/Decryption Engine** | *Verilog, FPGA, Cryptography* Source Code | *2023*
  - **Core Logic**: Synthesized robust AES cryptographic core in Verilog HDL supporting 128-bit, 192-bit, and 256-bit key standards.
  - **Datapath**: Engineered hardware-optimized modules for SubBytes, ShiftRows, MixColumns, and AddRoundKey transformations.
  - **Validation**: Verified cryptographic correctness against standard NIST test vectors using rigorous debug tracing.
- **Tactic (Robotics & AI)** | *Embedded Systems* Source Code | *2024*
  - **System Integration**: Developed browser-controlled robotics suite integrating AI computer vision with low-latency hardware control.
  - **Firmware**: Programmed ESP32 microcontrollers to handle WebSocket communication for real-time motor control and telemetry.
  - **Backend Processing**: Deployed YOLOv5 and OpenCV on host server to process visual feedback and automate robotic decision-making.

- **Tactic (Robotics & AI)** | *Real-Time Systems, WebSockets, Embedded* Source Code | *2024*
  - **Real-Time Messaging System**: Engineered a low-latency WebSocket communication layer to integrate a web dashboard with embedded hardware.
  - **Reliability**: Ensured reliable message delivery and system synchronization between the browser interface and the robotic unit.
- **Hankers (Social Platform)** | *Full-Stack Engineering* Source Code | *2025*
  - **Architecture**: Built scalable social platform using Microservices principles, React.js, and TypeScript.
  - **Frontend**: Implemented responsive UI design and asynchronous API integration via AJAX.
  - **DevOps**: Established CI/CD pipelines and containerized services using Docker for consistent deployment.
- **5-Stage Pipelined Processor (RISC)** | *VHDL, QuestaSim, Computer Architecture* Source Code | *2025*
  - **Core Architecture**: Engineered a 32-bit 5-stage pipelined processor in VHDL, supporting a custom ISA of over 25 instructions.
  - **Hazard Resolution**: Optimized throughput by implementing Hazard Detection and Data Forwarding units to resolve control and data dependencies.
  - **Verification**: Validated functional correctness through rigorous simulation and testbench environments in QuestaSim.
- **Hankers (Social Platform)** | *Microservices, TypeScript, Docker, CI/CD* Source Code | *2025*
  - **Scalable Microservices Architecture**: Designed a distributed system using Microservices and TypeScript to decouple services and ensure scalability.
  - **DevOps Automation**: Implemented CI/CD pipelines and Docker to automate testing and reliable delivery of services.
- **NexaMart** | *E-commerce Solution, Full-Stack* Source Code | *2025*
  - **Stack**: Next.js (TypeScript), Redux Toolkit, Laravel 11, and MySQL.
  - **Functionality**: Implemented secure authentication (Sanctum), shopping cart logic, real-time order processing, and administrative analytics dashboard.
- **Tayseer-Web** | *Corporate Web Solution, Next.js* Source Code | *2024*
  - **Deployment**: Deployed corporate website utilizing Next.js framework and TypeScript.
  - **Performance**: Maximized performance via Server-Side Rendering (SSR) and strict static typing for reliability.
- **Company Manager App – Al Nada Scientific Office** | *ERP System, React & PostgreSQL*Source Code | *2024*
  - **Tech Stack**: React.js, Redux, Express.js, Prisma, PostgreSQL, Electron for desktop deployment.
  - **Business Solution**: Developed ERP application for Al Nada Scientific Office managing customers, suppliers, inventory, and sales records.
  - **Deployment**: Packaged application with Electron for local server deployment, enabling offline usage.
- **Blog Web Application** | *Content Management System, Node.js* Source Code | *2024*
  - **Core Features**: Architected content platform with CRUD capabilities, rich-text editing (Quill), and secure image handling (Multer).
- **Interactive Orrery – NASA Space Apps Hackathon** | *3D Space Simulation, React.js*Source Code | *Oct 2024*
  - **Hackathon Project**: Developed during NASA Space Apps Challenge 2024, simulating orbital mechanics and near-Earth object trajectories using React.js and Three.js.
- **x86 Brick Breakers Arcade** | *Assembly (x86), MASM, Low-Level Programming* Source Code | *2024*
  - **Game Logic**: Developed a real-time, multiplayer arcade game using x86 Assembly and the MASM assembler.
  - **System Control**: Directly manipulated video memory for low-latency graphics rendering and managed hardware interrupts to handle real-time I/O.
- **Alien Invasion** | *Combat Simulation, C++* Source Code | *2023*
  - **Simulation**: Modeled combat scenarios using custom data structures to simulate fleet mechanics and collision detection.
- **Paint For Kids** | *C++, OOP, Data Structures* Source Code | *2023*
  - **Application**: Built desktop drawing application with shape tools, color fills, and interactive game modes.
  - **Design**: Applied OOP principles using polymorphism for extensible shape classes and Stack-based Undo/Redo system.

Technical Skills

- **Digital Design & Verification**: Verilog HDL, VHDL, RTL Design, RISC-V ISA, MIPS Architecture, Pipeline Design, RTL-to-GDSII Flow, OpenLane2, DRC/LVS Verification, Static Timing Analysis, FSM Design
- **Development Tools & Platforms**: ModelSim, QuestaSim, Intel Quartus Prime, GTKWave, ESP32, AVR/Arduino
- **Programming Languages**: C/C++ (Advanced), Python, x86 Assembly, MIPS Assembly